AMERICAN INTERNATIONAL UNIVERSITY-BANGLADESH (AIUB)
Department of Computer Science
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CSC 2210: OBJECT ORIENTED PROGRAMMING 2

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Section: k
Group No: 11

Project Report On

Turf Lagbe – Turf Booking System

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CO2: Display and verify the mean of a real-life Project using the concepts of C# Graphical User Interface based environment with database integration to depict a desktop-based application.

Assessment Criteria	Not Attended/ Incorrect (0)	Inadequate (1-2)	Average (3)	Good (4)	Excellent (5)
Evaluation Criteria	**Evaluation Definition**				**Total =**
Requirement fulfillment	Properly demonstrate a real-life scenario-based project with proper functional requirement identification for the Object Oriented Programming project development activities.				
Validation	Ensuring the ability of students' proper demonstration on validation forms in their system in terms of dealing with the data.				

Verification	Identifying if the students can verify the system data along with proper functional requirements in terms of data flow.	

Table of Contents

1. INTRODUCTION 3
2. FEATURE LIST 3
   2.1 Admin 3
   2.2 Turf Manager 3
   2.3 Regular User 3
3. ER DIAGRAM 4
   3.1 Normalization 5
   3.2 Finalization 6
4. UML DIAGRAM.. 10
   4.1 Use Case Diagram 10
   4.2 Activity Diagram 11

5. Sql queries……………………………………9
6.Conclusion……………………………………14

1. INTRODUCTION

The project is about Turf Booking Management System named “Turf Lagbe”, where multiple turfs, time slots, users, reservations and payments data are kept and handled. This method makes it easier to manage a small football turf reservation in Dhaka, Bangladesh. Most grass facilities nowadays are dependent on phone calls and walk-in bookings leading to doublebooking, conflicts and poor record keeping. This application tackles these challenges by providing a centralized desktop-based booking platform with role based access for Admins, Turf Managers and Regular Users. The project was built using C# Windows Forms (.NET 8.0) with Microsoft SQL Server as the database backend using Microsoft.Data.Database communication using SqlClient (ADO.net).

2. FEATURE LIST

2.1 Admin

The admin has full control over the system through a tabbed interface with three sections. The All-Bookings tab displays every booking across all turfs with filtering by status (Pending, Confirmed, Cancelled) and a real-time revenue summary. The Manage Turfs tab allows adding new turf facilities or updating existing ones (name, area, address, field type, capacity, phone). The Manage Slots tab lets the admin select a turf and configure its time slots including slot name, start/end times, and pricing in BDT. All CRUD operations are reflected immediately in the database.

2.2 Turf Manager

Each Turf Manager is assigned to a specific turf facility through the TurfManagers junction table. Upon login, the system identifies their assigned turf and displays a dedicated dashboard showing only bookings for that turf. The Manager can filter bookings by status (Pending, Confirmed, Cancelled) and view a summary bar showing total bookings, confirmed count, cancelled count, and total confirmed revenue in BDT. This role provides operational oversight without system-wide administrative access.

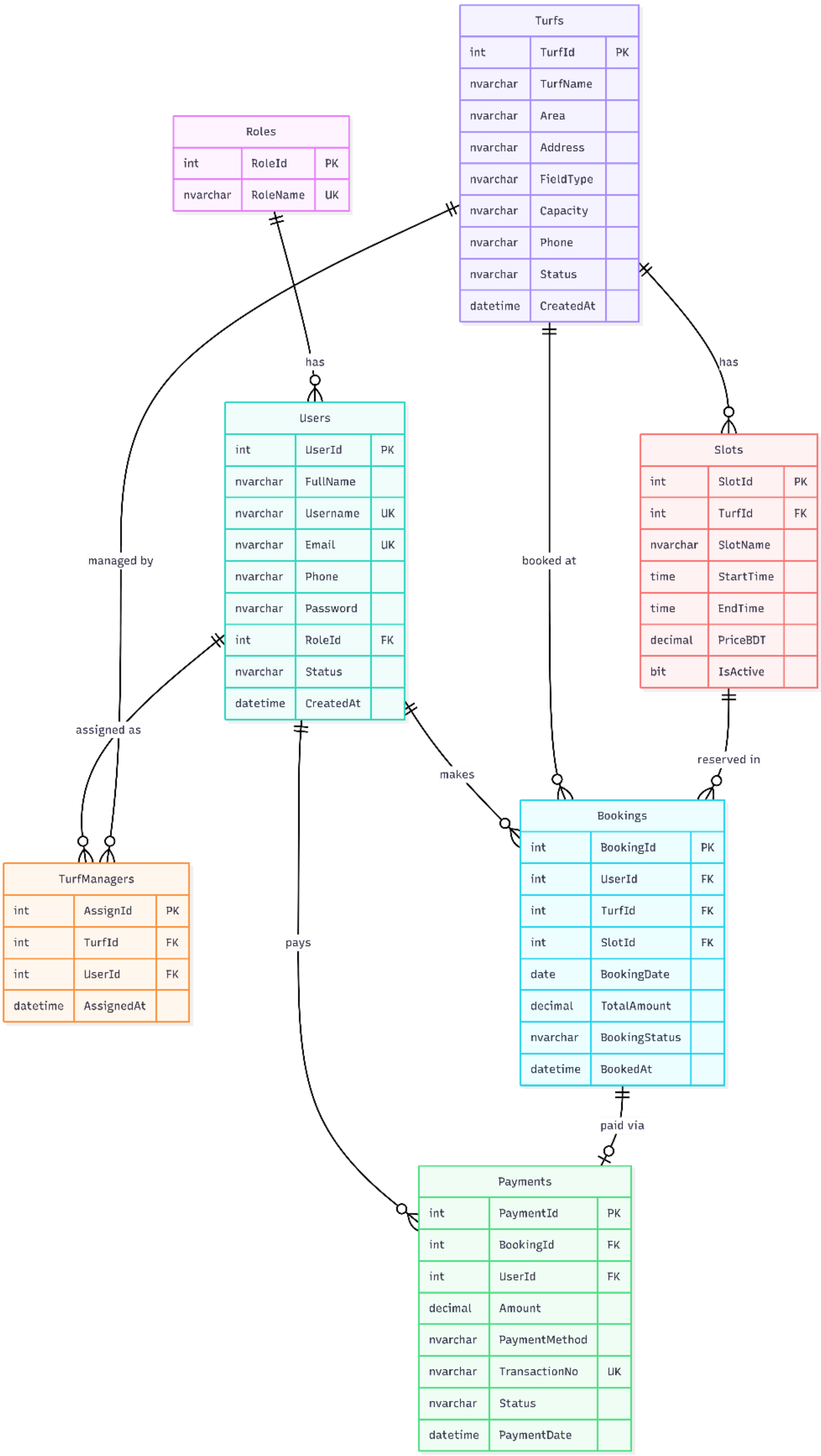
2.3 Regular User

Regular Users can browse all available turfs displayed as visual cards with turf images, location, field type, capacity, and price range. Clicking “Book Now” opens a slot selection form where the user picks a date using a DateTimePicker control. The system queries the database to show only available slots (excluding those already booked as Pending or Confirmed). After selecting a slot, the user confirms the booking which inserts records into both the Bookings and Payments tables. The "My Bookings" view displays all the user's bookings with the ability to cancel, which updates both the booking status to "Cancelled" and the payment status to "Refunded."

3. ER DIAGRAM

The Entity-Relationship diagram below shows seven entities and their relationships. The Roles table has a one-to-many relationship with Users. Users connect to TurfManagers, Bookings, and Payments. Turfs connect to TurfManagers, Slots, and Bookings. Slots connect to Bookings

, and Bookings connect to Payments.



3.1 Normalization

Bookings (Users-Turfs-Slots):

UNF: user_name, user_id, turf_name, turf_id, slot_name, slot_id, start_time, end_time, price, booking_date, total_amount, status

1NF: user_id(PK), user_name, turf_id(PK), turf_name, slot_id(PK), slot_name, start_time, end_time, price, booking_date, total_amount, status 2NF:

1) user_id(PK), user_name, email, phone, password, role_id(FK)
2) turf_id(PK), turf_name, area, address, field_type, capacity

3) slot_id(PK), slot_name, start_time, end_time, price, turf_id(FK)
4) booking_id(PK), user_id(FK), turf_id(FK), slot_id(FK), booking_date, total_amount, status

3NF: Same as 2NF — no transitive dependencies exist.

Payments (Bookings-Users):
UNF: booking_id, user_id, amount, payment_method, transaction_no, status, payment_date
1NF: payment_id(PK), booking_id, user_id, amount, payment_method, transaction_no, status, payment_date
2NF: payment_id(PK), booking_id(FK), user_id(FK), amount, payment_method, transaction_no(UK), status, payment_date
3NF: Same as 2NF — no transitive dependencies.

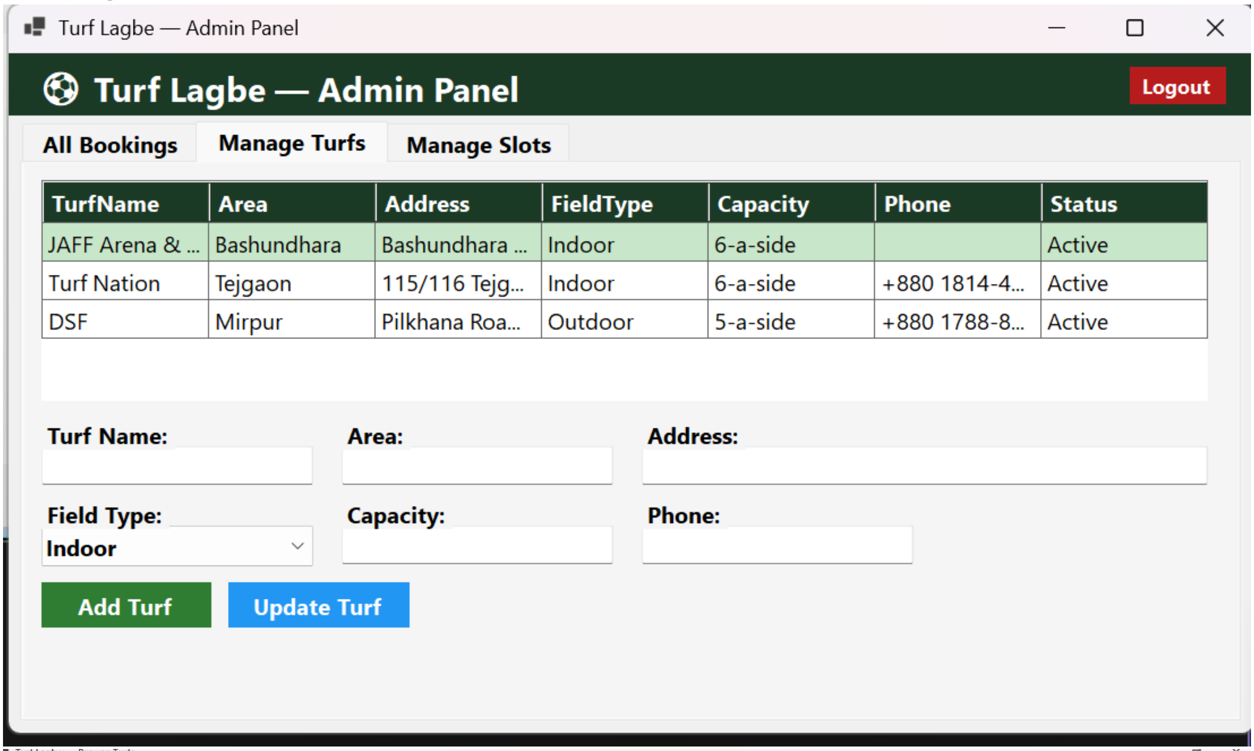
TurfManagers (Users-Turfs):
UNF: user_id, user_name, turf_id, turf_name, assigned_at
1NF: assign_id(PK), user_id, turf_id, assigned_at 2NF:
assign_id(PK), user_id(FK), turf_id(FK), assigned_at 3NF:
Same as 2NF.

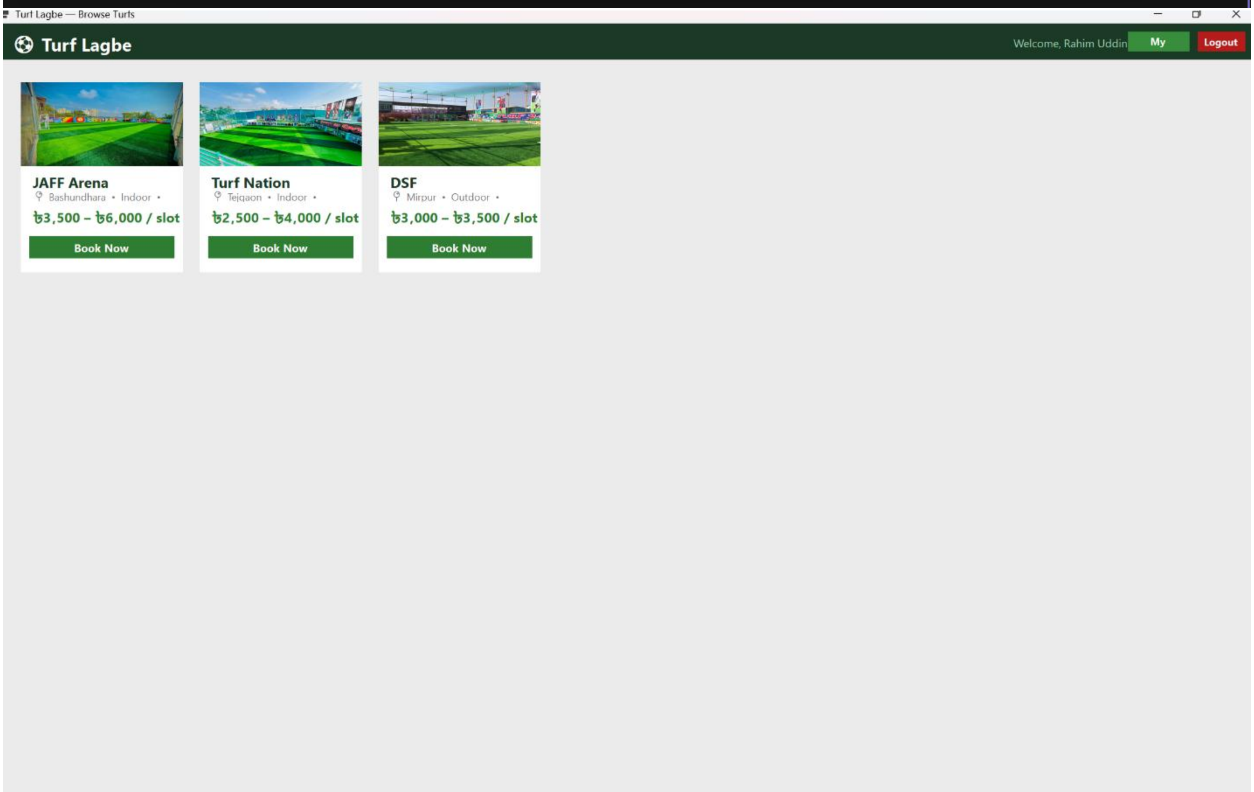
Roles-Users:
UNF: role_name, role_id, user_id, user_name, email
1NF: role_id(PK), role_name | user_id(PK), user_name, email, role_id 2NF:
1) role_id(PK), role_name(UK) 2) user_id(PK), full_name, username(UK), email(UK), phone, password, role_id(FK), status 3NF: Same as 2NF.

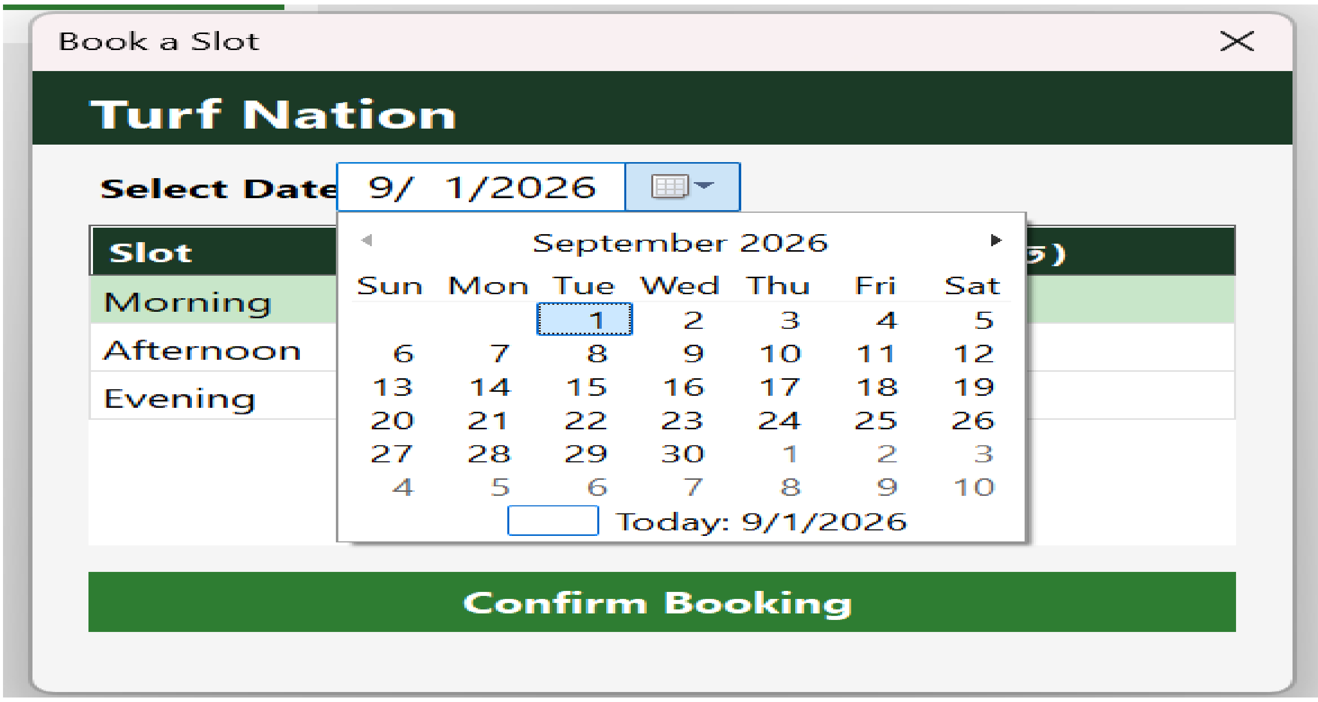
3.2 Finalization

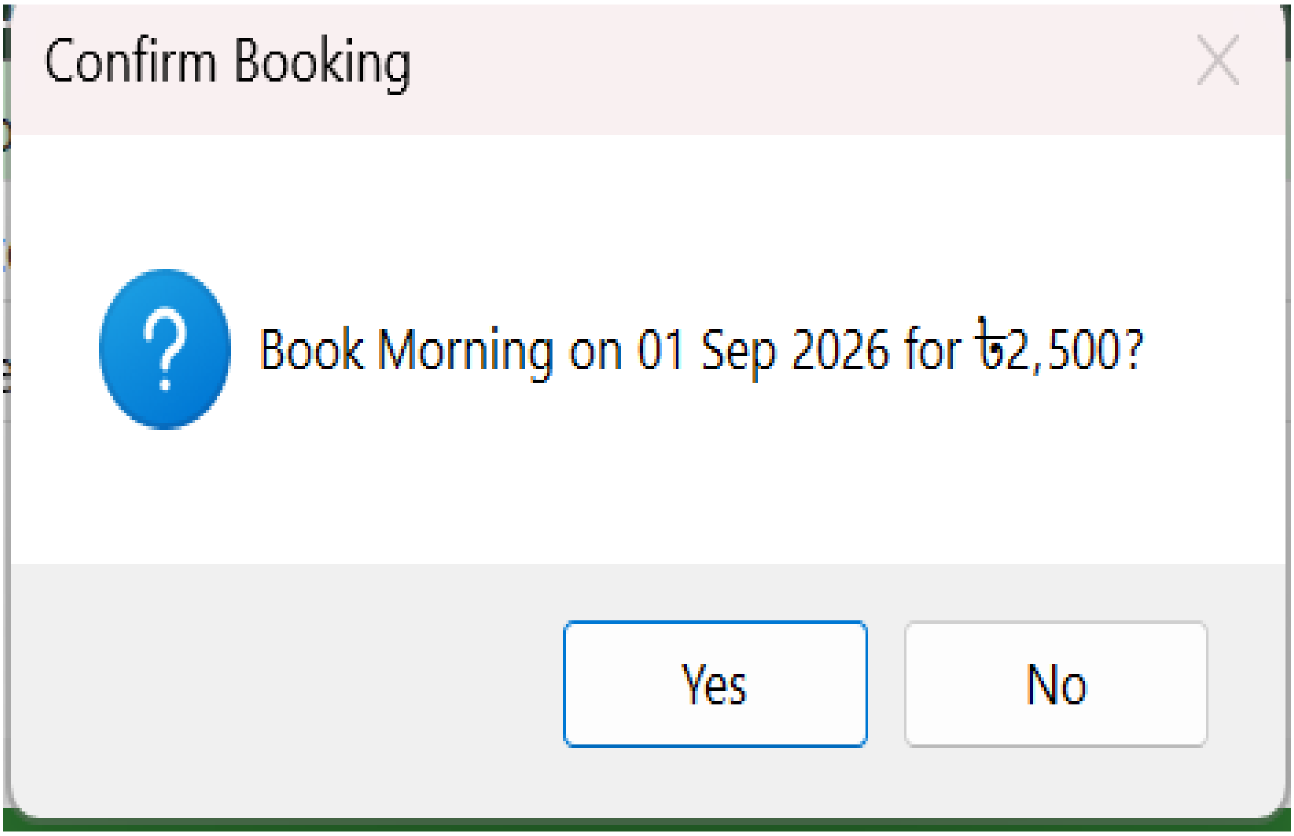
Final Tables (7 Tables):
1) Roles: role_id(PK), role_name(UK)
2) Users: user_id(PK), full_name, username(UK), email(UK), phone, password, role_id(FK), status, created_at
3) Turfs: turf_id(PK), turf_name, area, address, field_type, capacity, phone, status, created_at
4) TurfManagers: assign_id(PK), turf_id(FK), user_id(FK), assigned_at
5) Slots: slot_id(PK), turf_id(FK), slot_name, start_time, end_time, price_bdt, is_active
6) Bookings: booking_id(PK), user_id(FK), turf_id(FK), slot_id(FK), booking_date, total_amount, booking_status, booked_at
7) Payments: payment_id(PK), booking_id(FK), user_id(FK), amount, payment_method, transaction_no(UK), status, payment_date

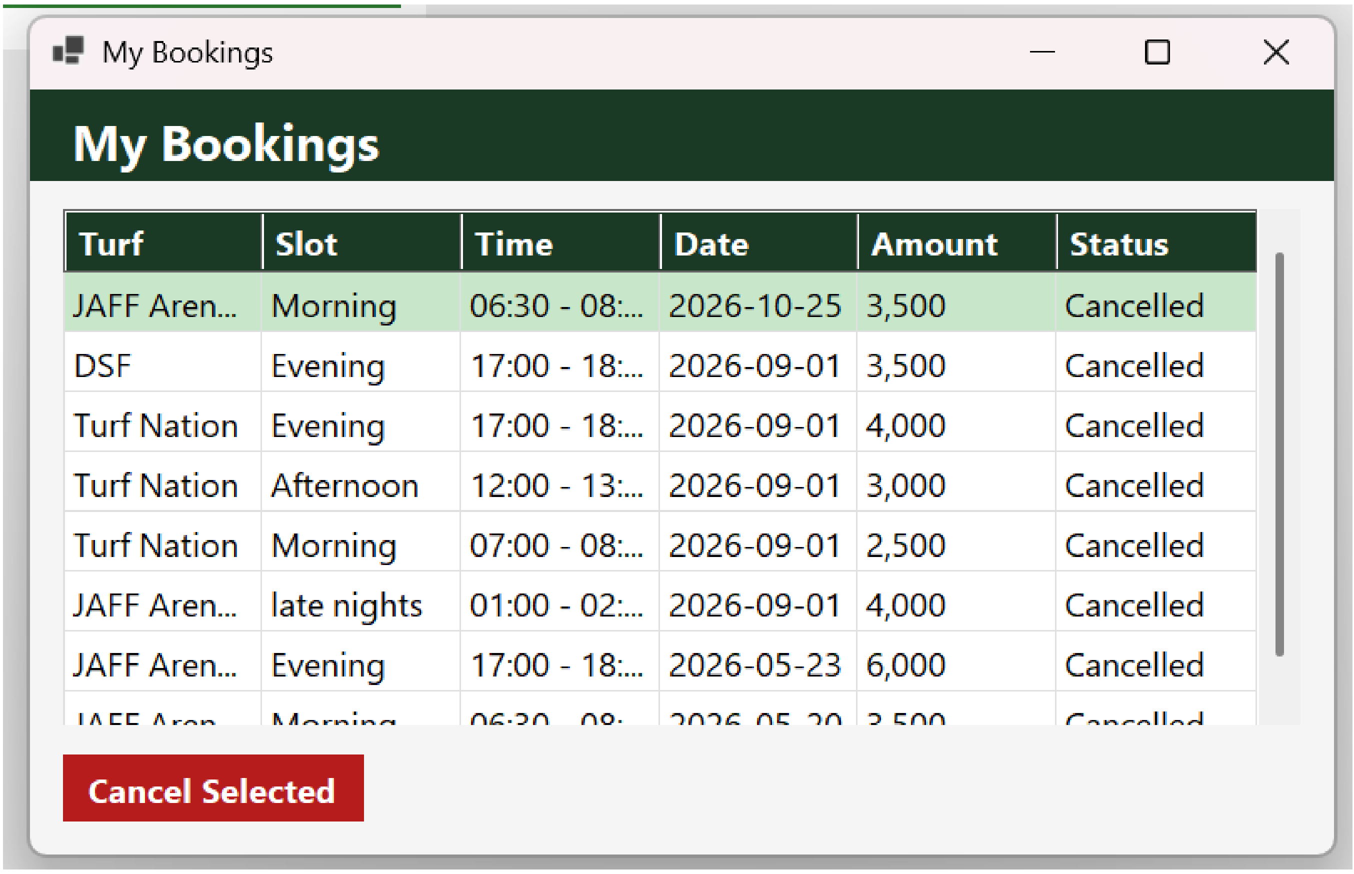
Design Forms:



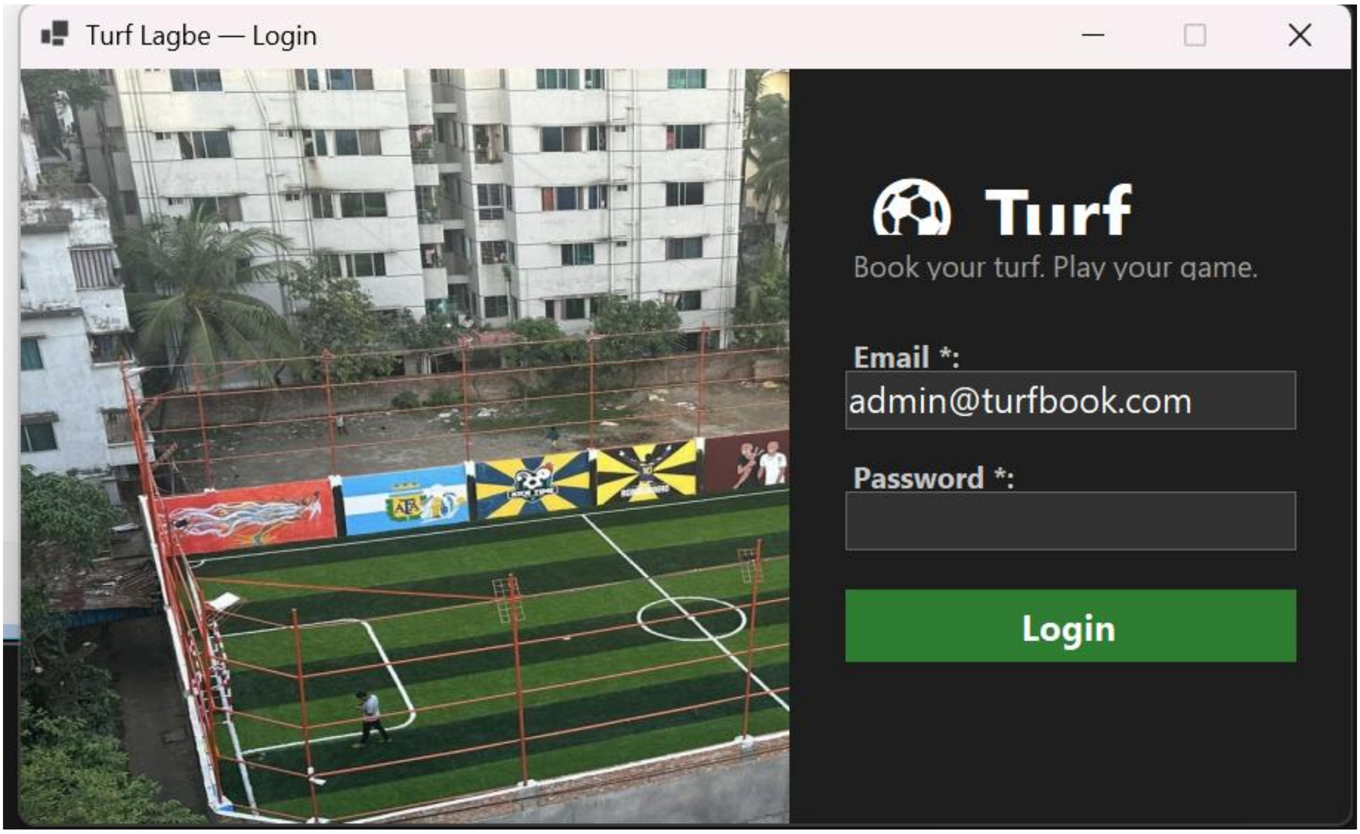



Book a Slot
Turf Nation
Select Date
9/ 1/2026
Slot
Morning
Afternoon
Evening
September 2026
Sun Mon Tue Wed Thu Fri Sat
Today: 9/1/2026
Confirm Booking


Confirm Booking
Book Morning on 01 Sep 2026 for ৳2,500?
Yes
No



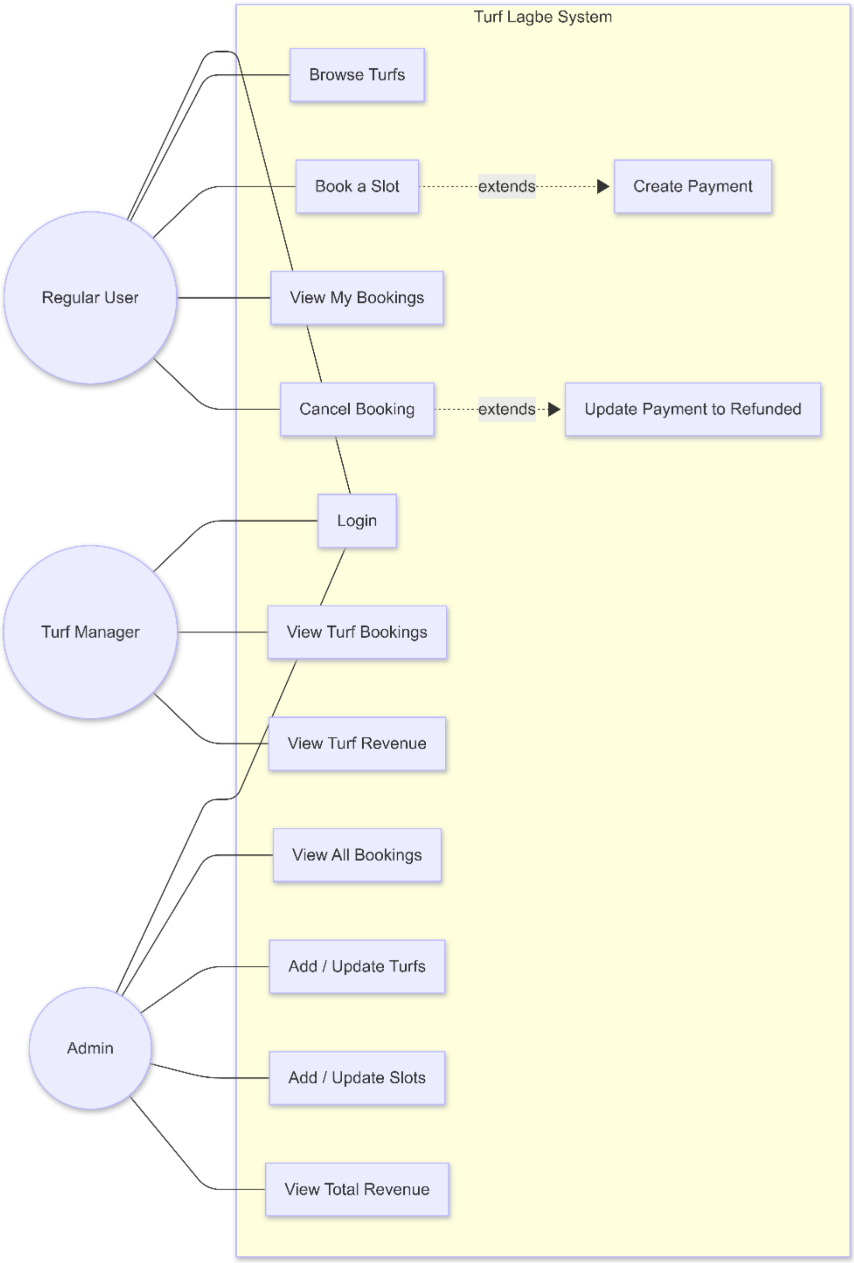
Turf	Slot	Time	Date	Amount	Status
JAFF Aren...	Morning	06:30 - 08:...	2026-10-25	3,500	Cancelled
DSF	Evening	17:00 - 18:...	2026-09-01	3,500	Cancelled
Turf Nation	Evening	17:00 - 18:...	2026-09-01	4,000	Cancelled
Turf Nation	Afternoon	12:00 - 13:...	2026-09-01	3,000	Cancelled
Turf Nation	Morning	07:00 - 08:...	2026-09-01	2,500	Cancelled
JAFF Aren...	late nights	01:00 - 02:...	2026-09-01	4,000	Cancelled
JAFF Aren...	Evening	17:00 - 18:...	2026-05-23	6,000	Cancelled



4. UML DIAGRAM

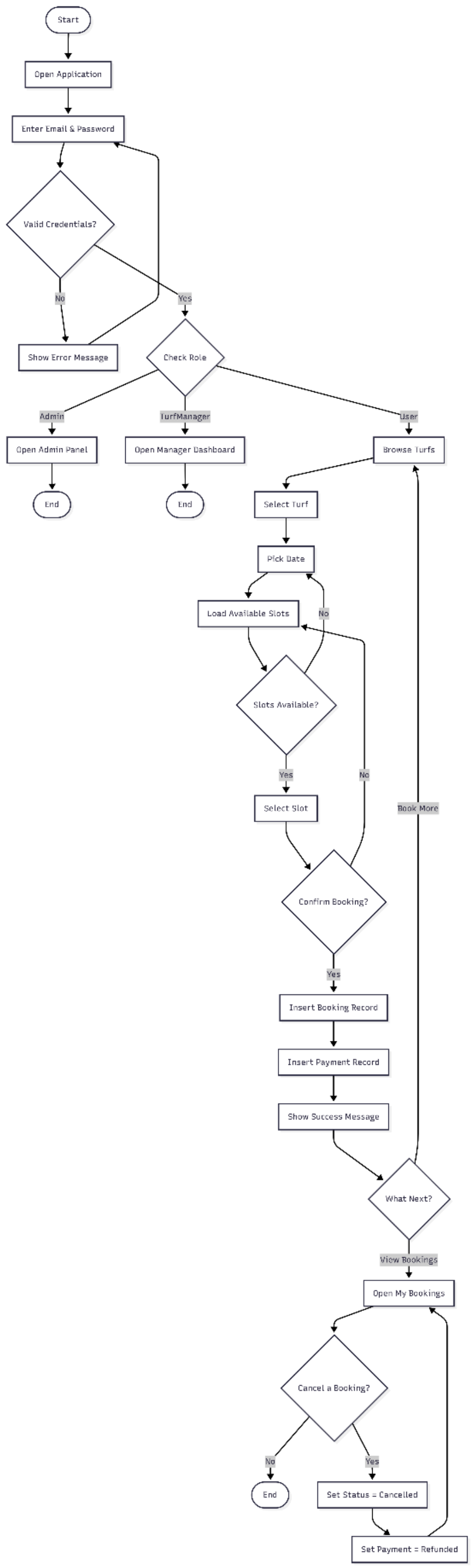
4.1 Use Case Diagram

In Use Case Diagram there are three actors, Admin, Turf Manager and Regular User. Admin may control turfs (Add/update) control slots (Add/update price and time) View all reservations with revenue The Turf Manager can see bookings and revenue for their turf. Regular User Login Browse Turfs Book slots View bookings Cancel bookings The Login use case is common to all actors and is role-based routing. Book Slot use case is extended to Create Payment and Cancel Booking is extended to Update Payment Status to Refunded.



4.2 Activity Diagram.

The Activity Diagram represents the whole booking process including launching the application, login, routing based on user roles, exploring the turf, slot selection, booking confirmation by creating a payment and flow of cancelation updating the refund status.



5. SQL Queries:

CREATE DATABASE TurfBookingSystem;

```sql
GO

USE TurfBookingSystem;
GO

-- ==========================================
-- ROLES
-- ==========================================

CREATE TABLE Roles (
   RoleId   INT PRIMARY KEY IDENTITY(1,1),
   RoleName NVARCHAR(50) UNIQUE NOT NULL
);

INSERT INTO Roles (RoleName) VALUES
('Admin'),
('TurfManager'),
('User');

-- ==========================================
-- USERS
-- ==========================================

CREATE TABLE Users (
   UserId    INT PRIMARY KEY IDENTITY(1,1),
   FullName  NVARCHAR(100) NOT NULL,
   Username  NVARCHAR(50)  UNIQUE NOT NULL,
   Email     NVARCHAR(100) UNIQUE NOT NULL,
   Phone     NVARCHAR(20),
   Password  NVARCHAR(255) NOT NULL,
   RoleId    INT NOT NULL,
   Status    NVARCHAR(20)  DEFAULT 'Active',
   CreatedAt DATETIME      DEFAULT GETDATE(),

   FOREIGN KEY (RoleId) REFERENCES Roles(RoleId)
);

INSERT INTO Users (FullName, Username, Email, Phone, Password, RoleId) VALUES
('System Admin',  'admin',   'admin@turfbook.com',    '01711000001', 'hashed_pw', 1),
('Karim Hossain', 'karim',   'karim@turfbook.com',    '01822000002', 'hashed_pw', 2),
('Rafi Ahmed',    'rafi',    'rafi@turfbook.com',     '01822000003', 'hashed_pw', 2),
('Sumon Islam',   'sumon',   'sumon@turfbook.com',    '01822000004', 'hashed_pw', 2),
('Rahim Uddin',   'rahim',   'rahim@gmail.com',       '01933000005', 'hashed_pw', 3),
('Nusrat Jahan',  'nusrat',  'nusrat@gmail.com',      '01933000006', 'hashed_pw', 3),
('Tanvir Khan',   'tanvir',  'tanvir@gmail.com',      '01933000007', 'hashed_pw', 3);

-- ==========================================
```

```
-- TURFS
-- ==========================================

CREATE TABLE Turfs (
   TurfId      INT PRIMARY KEY IDENTITY(1,1),
   TurfName    NVARCHAR(100) NOT NULL,
   Area        NVARCHAR(100),
   Address     NVARCHAR(MAX),
   FieldType   NVARCHAR(50),
   Capacity    NVARCHAR(50),
   Phone       NVARCHAR(20),
   Status      NVARCHAR(20) DEFAULT 'Active',
   CreatedAt   DATETIME     DEFAULT GETDATE()
);

INSERT INTO Turfs (TurfName, Area, Address, FieldType, Capacity, Phone) VALUES
('JAFF Arena & Academy', 'Bashundhara', 'Bashundhara Main Gate, Dhaka 1229',          'Indoor',  '6-a-side', NULL),
('Turf Nation',          'Tejgaon',     '115/116 Tejgaon Industrial Area, Dhaka 1208',  'Indoor',  '6-a-side', '+880 1814-460000'),
('DSF',                  'Hazaribagh',  'Pilkhana Road, Dhaka 1205',                    'Outdoor', '5-a-side', '+880 1788-857133');

-- ==========================================
-- TURF MANAGERS
-- ==========================================

CREATE TABLE TurfManagers (
   AssignId   INT PRIMARY KEY IDENTITY(1,1),
   TurfId     INT NOT NULL,
   UserId     INT NOT NULL,
   AssignedAt DATETIME DEFAULT GETDATE(),

   FOREIGN KEY (TurfId) REFERENCES Turfs(TurfId),
   FOREIGN KEY (UserId) REFERENCES Users(UserId)
);

INSERT INTO TurfManagers (TurfId, UserId) VALUES
(1, 2),   -- Karim  -> JAFF Arena
(2, 3),   -- Rafi   -> Turf Nation
(3, 4);   -- Sumon  -> DSF

-- ==========================================
-- SLOTS
-- ==========================================

CREATE TABLE Slots (
```

```sql
    SlotId      INT PRIMARY KEY IDENTITY(1,1),
    TurfId      INT NOT NULL,
    SlotName    NVARCHAR(50)  NOT NULL,
    StartTime   TIME          NOT NULL,
    EndTime     TIME          NOT NULL,
    PriceBDT    DECIMAL(10,2) NOT NULL,
    IsActive    BIT           DEFAULT 1,

    FOREIGN KEY (TurfId) REFERENCES Turfs(TurfId)
);

-- JAFF Arena (TurfId = 1)
INSERT INTO Slots (TurfId, SlotName, StartTime, EndTime, PriceBDT) VALUES
(1, 'Morning',   '06:30', '08:00', 3500),
(1, 'Afternoon', '13:00', '14:30', 4500),
(1, 'Evening',   '17:00', '18:30', 6000),
(1, 'Night',     '20:00', '21:30', 6000);

-- Turf Nation (TurfId = 2)
INSERT INTO Slots (TurfId, SlotName, StartTime, EndTime, PriceBDT) VALUES
(2, 'Morning',   '07:00', '08:30', 2500),
(2, 'Afternoon', '12:00', '13:30', 3000),
(2, 'Evening',   '17:00', '18:30', 4000);

-- DSF (TurfId = 3)
INSERT INTO Slots (TurfId, SlotName, StartTime, EndTime, PriceBDT) VALUES
(3, 'Morning', '07:30', '09:00', 3000),
(3, 'Evening', '17:00', '18:30', 3500);

-- ==========================================
-- BOOKINGS
-- ==========================================

CREATE TABLE Bookings (
    BookingId     INT PRIMARY KEY IDENTITY(1,1),
    UserId        INT NOT NULL,
    TurfId        INT NOT NULL,
    SlotId        INT NOT NULL,
    BookingDate   DATE          NOT NULL,
    TotalAmount   DECIMAL(10,2) NOT NULL,
    BookingStatus NVARCHAR(30)  DEFAULT 'Pending',
    BookedAt      DATETIME      DEFAULT GETDATE(),

    FOREIGN KEY (UserId) REFERENCES Users(UserId),
    FOREIGN KEY (TurfId) REFERENCES Turfs(TurfId),
    FOREIGN KEY (SlotId) REFERENCES Slots(SlotId)
);
```

```sql
INSERT INTO Bookings (UserId, TurfId, SlotId, BookingDate, TotalAmount, BookingStatus) VALUES
(5, 1, 1, '2026-05-20', 3500, 'Confirmed'),
(6, 2, 5, '2026-05-21', 2500, 'Confirmed'),
(7, 3, 8, '2026-05-22', 3000, 'Pending'),
(5, 1, 3, '2026-05-23', 6000, 'Cancelled');

-- ============================================
-- PAYMENTS
-- ============================================

CREATE TABLE Payments (
    PaymentId     INT PRIMARY KEY IDENTITY(1,1),
    BookingId     INT NOT NULL,
    UserId        INT NOT NULL,
    Amount        DECIMAL(10,2) NOT NULL,
    PaymentMethod NVARCHAR(50),
    TransactionNo NVARCHAR(100) UNIQUE,
    Status        NVARCHAR(30)  DEFAULT 'Pending',
    PaymentDate   DATETIME      DEFAULT GETDATE(),

    FOREIGN KEY (BookingId) REFERENCES Bookings(BookingId),
    FOREIGN KEY (UserId)    REFERENCES Users(UserId)
);

INSERT INTO Payments (BookingId, UserId, Amount, PaymentMethod, TransactionNo, Status) VALUES
(1, 5, 3500, 'bKash', 'TXN-BK-001', 'Completed'),
(2, 6, 2500, 'Nagad', 'TXN-NG-002', 'Completed'),
(3, 7, 3000, 'bKash', 'TXN-BK-003', 'Pending');

-- ============================================
-- PREVENT DOUBLE BOOKING (Optional Safety)
-- ============================================

CREATE UNIQUE INDEX UX_ActiveBooking
ON Bookings(SlotId, BookingDate)
WHERE BookingStatus IN ('Pending', 'Confirmed');
GO
```

6. CONCLUSION

This project was on Turf Booking Management System dubbed "Turf Lagbe" constructed using C# Windows Forms (.NET 8.0) and Microsoft SQL Server. Firstly, a case study was created outlining the business scenario, and then an ER diagram was designed displaying seven entities and their relationships. The ER diagram was normalized from UNF to 3NF and the final tables were constructed in SQL Server using CREATE TABLE commands, and sample data was entered. The main OOP ideas used are encapsulation (shared state is stored in AppData and Session classes), polymorphism (turf cards are dynamically generated for the UI), and modularity (each form is responsible for one thing). Several queries were tested: simple queries, aggregate functions, subqueries, joins and views. Created UML Diagrams (Use Case, Activity)